Settings Reference

Every field on the navigation manager, with verified defaults and limits.

Every field on the navigation manager, in the order the Details panel presents them. Defaults and limits below are the shipping values.

If you only read one thing: the four settings that decide whether the plugin works well in your level are `Infinite World`, `World Extent` / `Chunk Size`, `Min Voxel Size`, and `Clearance Padding`. Everything else is refinement.

Concepts behind these are explained in Tutorial 02: Core Concepts.

1. Setup

`Infinite World`: default **off**

The biggest decision. Off = one fixed navigation volume built up front. On = the world is divided into chunks generated around agents as they move.

Leave it off unless you have a genuinely unbounded or streaming world. Finite is simpler, faster and more predictable.

Ticking this hides `World Extent` and reveals `Chunk Size` and the skeleton options.

`Auto Build On Play`: default **on** *(finite only)*

Builds the navigation volume at `BeginPlay`. Turn it off only if you want to trigger `BuildSVO` yourself from Blueprint at a specific moment.

`Generation Strategy`: read-only *(infinite only)*

Shows which streaming strategy was selected automatically.

`Persistent Skeleton (runtime, no bake)`: default **on** *(infinite only)*

Keeps a lightweight long-range connectivity graph in memory so distant routes can be planned before the detailed chunks exist. **Leave this on**. Without it, long routes in an infinite world fail until the intervening chunks stream in.

`Use Baked Skeleton (frame-0 completeness)`: default off *(infinite only)*

Loads a pre-baked skeleton so long-range routing is complete from the very first frame. Requires running **Bake Skeleton (Editor)** first. Useful when agents must plan across the map immediately at level start.

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2. Voxels

`World Extent`: default 10000 cm *(finite only)*

Half-width of the navigation volume, measured outward from the manager actor. The default gives a 200 m cube.

Range: 500-20000. Recommended UI range: 2000-20000.

> **NOTE**
>
> **Anything outside this box has no navigation data.** Paths to targets outside it fail. Size it to cover your play space, and place the manager at its centre.

`Chunk Size`: default 8000 cm *(infinite only)*

Edge length of one streaming chunk.

Range: 1000-40000. Recommended UI range: 3000-8000.

> **WATCH OUT**
>
> **Do not treat this as a free dial.** It affects memory, generation cost, *and* whether long routes can be found at all. **8000 is the tested default.** If you change it, re-test long-range routing specifically: a value that looks fine for short hops can break distant paths.

`Min Voxel Size`: default 100 cm

The finest resolution the octree will subdivide to. **The dominant quality/cost control.**

Range: 10-2000. Recommended UI range: 25-500.

- Smaller → finds tighter gaps; more memory; longer builds.
- Larger → cheaper; narrow openings disappear.

Rule of thumb: about **half the narrowest gap** agents must pass through.

> **NOTE**
>
> **It's a floor, not an exact size.** The octree halves down from the world/chunk size, so effective resolution snaps to powers of two. 100 → 90 may change nothing; 100 → 60 may double memory. Tune by testing.

`Clearance Padding`: default 100 cm

Inflates obstacles during voxelization so agents keep clear of surfaces.

Range: 0-1000. Recommended UI range: 0-300.

> **WATCH OUT**
>
> **Keep this well below `Min Voxel Size`.** Padding that approaches voxel size fills in doorways and corridors completely, and the pathfinder reports no route through openings you can clearly see. **This is the single most common cause of "it says there's no path but I can see the gap".**

`Coarse Cell Size`: default 2500 cm

Cell size of the coarse long-range routing layer. Larger = cheaper long-range planning, less precise. The default suits most levels.

Range: 100-20000.

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3. Collision

`Obstacle Query Channels`

Which collision channels count as obstacles. Anything on these channels is voxelized as solid.

> **NOTE**
>
> If a wall isn't blocking agents, check that its collision channel is in this list **and** that the mesh actually has collision enabled.

`Allowed Obstacle Classes`

Optional filter. Leave **empty** to consider all actors (normal). Populate it to restrict voxelization to specific actor classes (useful when you want to ignore scenery entirely).

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4. Performance

`Use GPU Voxelization`: default on

Builds the octree on the GPU, which is substantially faster. Falls back to CPU automatically if no suitable GPU is present. Turn it off only to force deterministic, reproducible builds.

`Max Generations Per Frame`: default 1

Chunk builds allowed per frame. Higher fills the world faster but can cause hitches. *Range: 1+, UI up to 8.*

`Max Path Requests Per Frame` : default 64

Path queries dequeued per frame. Raise for many agents replanning at once; lower to protect frame time. *Range: 1-256.*

`Max Requests Per Frame` : default 50

General request throttle. *Range: 1-200.*

`Max Flow Solves Per Frame` : default 4

Flow-field solves per frame. *Range: 0-16.*

`Finite Obstacle Update Interval` : default 0.1 s *(finite only)*

How often dynamic obstacle changes are folded into the octree. Lower = more reactive, more CPU. *Range: 0-2 s.* See Tutorial 05.

`Chunk Quantized Macro Cache` : default off

Caches long-range routes quantized to chunk boundaries. Helps when many agents share similar long-range routes (large crowds spawning together).

Pawn Agents (BT & StateTree) subcategory

Field	Default	Meaning
`Use Flow Fields (pawn agents only)`	off	Opt-in: pawn agents heading to a shared goal use one steering field instead of individual paths
`Flow Field Threshold`	`16`	Agent count at which a shared field is built *(range 2-512)*
`Flow Field Cluster Cell Size`	`1000` cm	Grouping granularity for shared goals *(range 100-10000)*

> **NOTE** Flow fields are for **swarms heading to the same place**. For a handful of agents with individual goals they are pure overhead, so leave it off.

5. Debug

`Draw Debug Volumes` : default off

Draws the octree. **The most useful diagnostic in the plugin**: if a gap shows no fine voxels, the pathfinder can't route through it either. Expensive; turn it off when you're done.

Also available from Blueprint via `SetDrawDebugVolumes` / `ToggleDrawDebugVolumes`.

`Draw Pathfinding Debug`: default off

Draws computed paths.

`Draw Infinite World Debug`: default off *(infinite only)*

Draws chunk boundaries and streaming state.

`Force Redraw SVO` / `Debug Redraw Coalesce Seconds`: default 0.25 s

Forces a redraw, and how long redraws are batched. Raise the coalesce value if debug drawing itself is hurting frame rate.

6. Runtime State

Read-only diagnostics showing live counts of resident chunks, skeleton state and staleness. Useful when tuning an infinite world; nothing to set.

Build & Tools (Blueprint-callable)

Function	Purpose
`BuildSVO`	Build the navigation volume now
`BuildSVOInEditor`	Build in-editor, so EQS previews work without playing
`BakeSkeletonInEditor`	Pre-bake the long-range skeleton for `Use Baked Skeleton`
`LoadBakedSkeleton`	Load a previously baked skeleton
`FindPath`	Synchronous path query: start, end, options, out array of points
`FindCover`	Tactical cover query
`UpdateDynamicObstacle`	Manually stamp an obstacle by bounds
`SetMacroTopologyCell`	Manually mark a coarse cell blocked/free
`ForceRedrawSVO`, `SetDrawDebugVolumes`, `ToggleDrawDebugVolumes`, `DebugDrawFlowField`	Debug

Per-agent settings

These live on the `LCM Fly To (SVO)` BT task / **Fly To** StateTree task, not the manager:

Field	Default	Meaning
`Acceptance Radius`	`100` cm	Distance at which the goal counts as reached
`Flight Speed`	`600` cm/s	Cruise speed; match your movement component
`Flight Style`	*Physics (Drones/Ships)*	vs *Linear (Biological/Magic)*
`Avoidance Style`	*Reactive (Boids/Push)*	None / Reactive / Predictive (ORCA) / Context-Based
`Separation Radius`	`150` cm	Neighbour spacing
`Separation Weight`	`2.0`	Spacing strength
`Max Acceleration`	`2000` cm/s^2	Lower = heavier feel
`Banking Amount`	`4.0`	Roll into turns
`Max Banking Angle`	`45°`	Roll limit
`Deceleration Distance`	`800` cm	Slow-down lead-in
`Avoidance Time Horizon`	`2.0` s	ORCA look-ahead (Predictive only)
`Enable Goal Prediction`	off	Intercept a moving target
`Lead Seconds Cap`	`2.0` s	Prediction limit
`Drift Replan Cm`	`400` cm	Target movement before replanning
`Assist Infinite Streaming`	off	Helps agents cross not-yet-generated chunks in infinite worlds

Solver and smoothing choices are described in Tutorial 02 §6.

Recommended starting points

Scenario	`Infinite`	Extent / Chunk	`Min Voxel Size`	`Clearance Padding`
Arena or single building	off	`10000`	`100`	`100`
Tight interiors, small agents	off	`10000`	`50`	`25`
Large outdoor level	off	`20000`	`200`	`100`
Open / streaming world	**on**	`8000`	`100`	`100`

Start here, then confirm with `Draw Debug Volumes` that the gaps you care about actually have voxels in them.

Chapter 13 of the LCM Nav3D User Manual.
The complete manual is `LCM_Nav3D_Manual.pdf`, in this folder alongside every other document in the set. Start from `README.pdf`.